

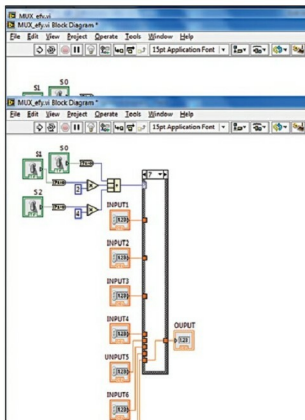


Fig. 3: Block diagram of 8-to-1 MUX after pressing Ctrl+U keys

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com

labels in the box appear as true and false, modify the cases to 0 and 1, respectively. Right click on Case Structure and select Add Case After option and label it as 2. Similarly, add five more such cases and label them sequentially as 3, 4, 5, 6 and 7.

7. Right click on block diagram → Programming → Boolean → Bool to (0, 1) and drag and drop on to block diagram and name it as S0. Similarly, do it for S1 and S2.

8. Right click on block diagram → Programming → Numeric → Multiply and drag and drop it on to block diagram. Place the cursor at output of Bool to (0, 1) of S1 and you will see a hot spot. Connect a wire from the hot spot to the input of the multiplier. Right click on the multiplier and create a constant of value 2. Repeat the same step for S2 with a constant of 4 to the multiplier.

9. Click on block diagram → Programming → Numeric → Compound

Arithmetic, drag and drop it on to the block diagram. Extend the Compound Arithmetic block using mouse such that it has three inputs. Give two inputs (S1 and S2) to Compound Arithmetic block via multipliers and another input from Bool to (0, 1) of S0. Right click Compound Arithmetic, select Change Mode → Add option.

10. Give the output of the Compound Arithmetic to '?' terminal on the Case Structure.

11. Now select case 0 and connect input 1 to output through the Case Structure. An open box called Tunnel will be created on the right side of the Case Structure. Select different cases (1 through 7) and connect inputs through this tunnel.

12. After all the inputs are given for all the cases (0 through 7), the tunnel will be shaded. Go to case 0 and right click on the Case Structure and select 'Make this default case' option. This completes the required connections of block diagram.

13. Wire connections of block diagram are shown in Fig. 2. If any error occurs during connection, press Ctrl+B from the keyboard to remove broken wires, or right click on the wire and select 'Delete Wire Branch' option to remove the wire. After completion of the wire connections, press Ctrl+U keys to clean up the diagram as shown in Fig. 3.

14. Return to the front panel and save the VI project. Click Run from Operate menu option to check the operation of 8-to-1 multiplexer. For example, if all S0, S1 and S2 are high (by pulling them up), only input 8 will be available at the output. Similarly, you can check the output by toggling the switches S0, S1 and S2 with different combinations. ●

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